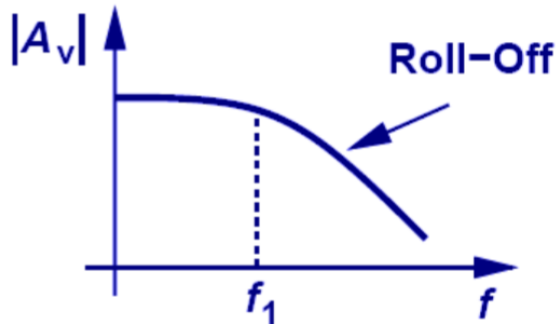


Frequency Response

High Frequency Rolloff

- As frequency increases, the gain of an amp generally decreases



-3 dB Bandwidth

- At corner frequency $|H(j\omega_c)| = \frac{A_0}{\sqrt{2}}$
- Example: $|A_v| = \frac{g_m R_D}{\sqrt{R_D^2 C_L^2 \omega^2 + 1}} = \frac{g_m R_D}{\sqrt{2}}$

- Power becomes half at corner frequency

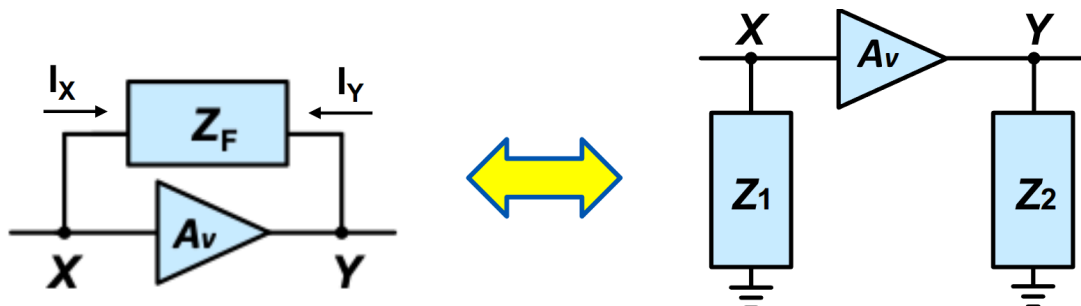
Bode Plot

- As ω passes each pole frequency (denominator, slope decreases by 20 dB/dec)
- As ω passes each zero frequency (numerator, slope increases by 20 dB/dec)
- Form to identify poles and zeros:

$$10 \frac{1 + s/10^3}{(1 + s/10^2)(1 + s/10^4)(1 + s/10^5)}$$

Intuitive Pole Identification

- If a capacitor terminal is ground, it creates a pole
- if capacitor acts as feedback in amp, then



where $Z_1 = \frac{Z_f}{1-A_v}$ and where $Z_2 = \frac{Z_f}{1-A_v^{-1}}$

- Simplifying

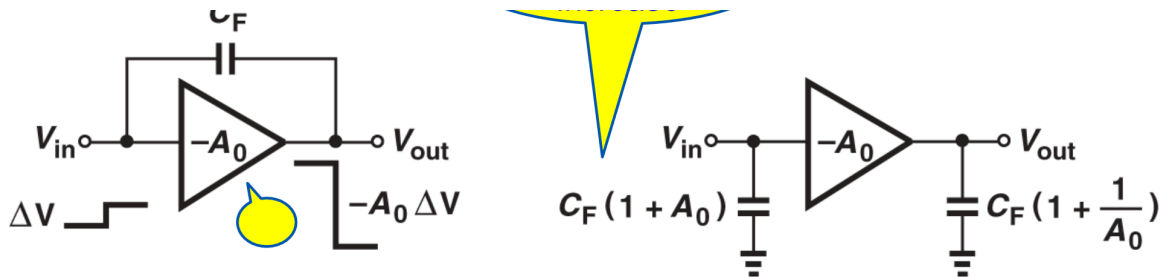
$$Z_1 = (1 - A_v)C_f \cdot s$$

$$Z_2 \approx C_f \text{ for large } |A_v|.$$

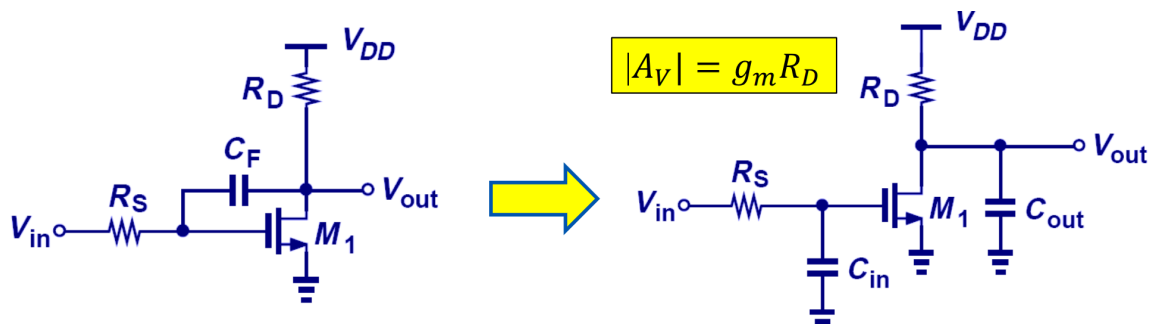
- Want small caps to increase bandwidth

Miller Effect

- if gain is A_0 , input capacitance for the below config increases by $1 + A_0$



- For example



where

$$C_{in} = (1 + A_0)C_F$$

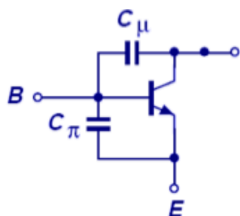
$$= (1 + g_m R_D)C_F$$

$$C_{out} = \left(1 + \frac{1}{g_m R_D}\right)C_F$$

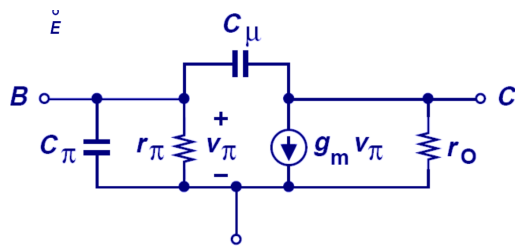
High Frequency Models

BJT

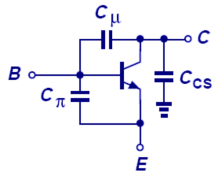
- parasitic capacitance between base and collector is denoted C_μ
- capacitance between base and emitter is C_π , is equal to $C_b + C_{je}$



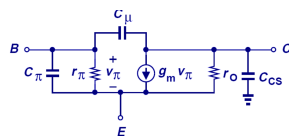
- In small signal,



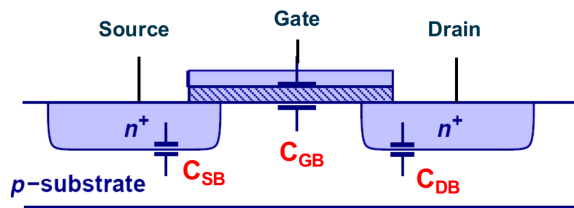
- In the IC BJT model, there is another parasitic capacitance from collector to ground denoted C_{cs}



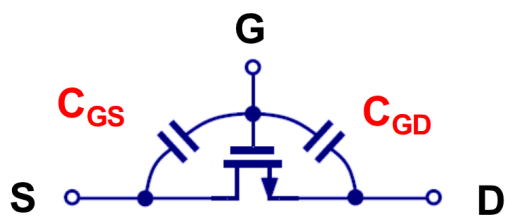
- in small signal,



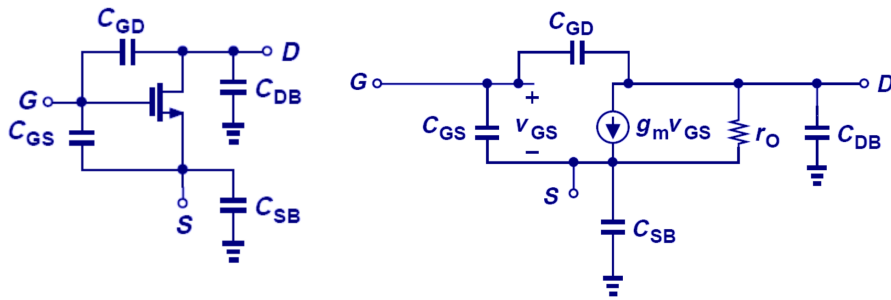
MOSFET



- between gate and drain:



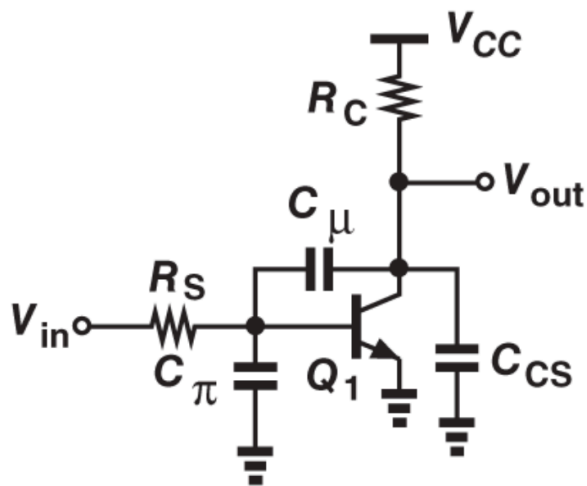
- Circuit model for MOSFET:



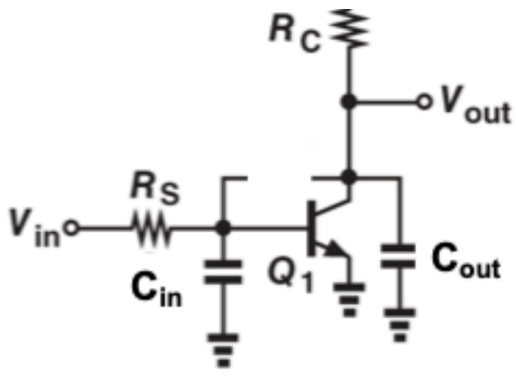
- Capacitance depends on operating conditions of MOSFET
 - E.g. C_{gs} is largest in saturation region

CE and CS Amps (Frequency Dependent Model)

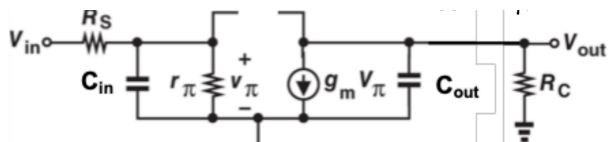
CE Amp



- Feedback cap C_μ makes finding A_v complicated
 - Use Miller's theorem, separate capacitance into C_{in} and C_{out} e.g.



- This makes the small signal model



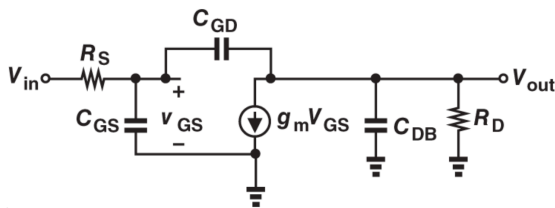
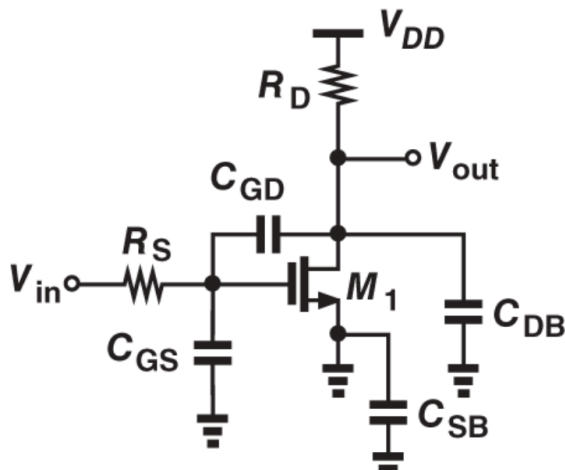
where

$$C_{in} = C_{\pi} + (1 + g_m R_C) C_{\mu} \quad C_{out} = C_{CS} + \left(1 + \frac{1}{g_m R_C}\right) C_{\mu}$$

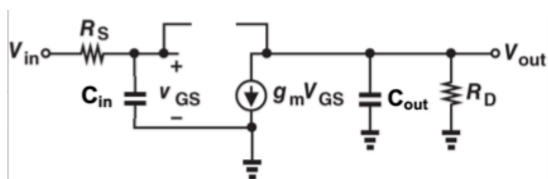
$$\approx C_{CS} + C_{\mu}$$

- 2 poles
- Remember that gain · bandwidth = k where k is some constant
 - As gain increases, bandwidth decreases, and vice versa

CS Amp



- Same process, use Miller's theorem to remove feedback cap, replace with equivalent model
 - Now, the small signal model is



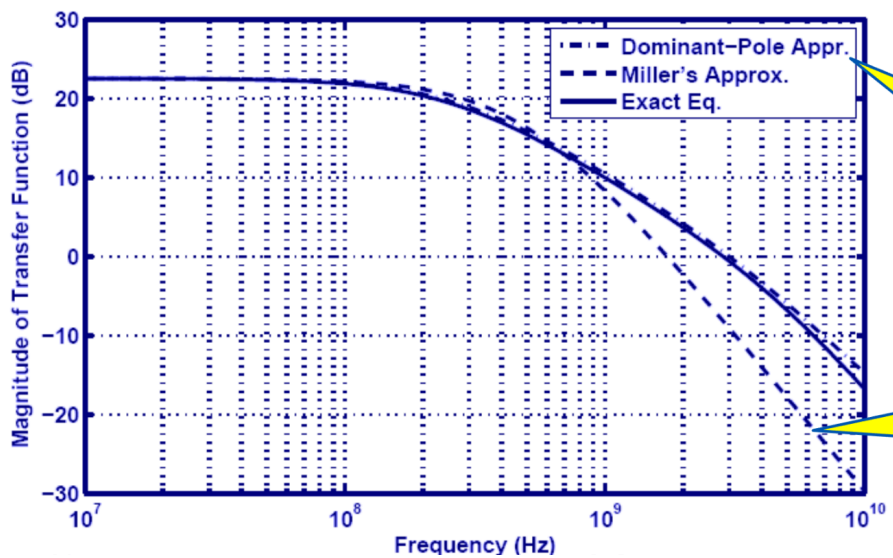
which is easier to analyze

- 2 poles

Bandwidth

- Miller effect increases C_{in} by a factor relating to gain
 - As gain increases, C_{in} increases, and bandwidth decreases, larger C_{in} results in lower frequency pole
 - The input pole caused by C_{in} is dominant over pole caused by C_{out}

- Miller's theorem is an approximation



- Approximation diverges from reality as frequency increases

Input Impedance of CE Amp

$$Z_{in} \approx \frac{1}{[C_{\pi} + (1 + g_m R_C)C_{\mu}]s} \parallel r_{\pi}$$

- ,
- As frequency increases, Z_{in} decreases

Input Impedance of CS Amp

$$Z_{in} \approx \frac{1}{[C_{GS} + (1 + g_m R_D)C_{GD}]s}$$

- As frequency increases, Z_{in} decreases